Fullerene chemistry continues to be an exciting field generating many articles with promising w applications every year. Some examples are given below:

(*i*) Magnetic nanoparticles show great potential for *high-density magnetic storage media.*

(*ii*) C-60 dispersed into ferromagnetic materials such as iron, cobalt can form thin films with promising *magnetic properties.*

(*iii*) A number of *organometallic-fullerene compounds* have recently been synthesized which are being used *in industry.*

(*iv*) Some fullerene derivatives even exhibit *superconducting character.* There has been a report of a fullerene containing, superconducting field-effect device with a T_c as high as 117 K.

(*v*) Fullerenes are used as miniature *ball bearings for lubrication.*

(*vi*) They can be used *as drug deliverers* by the attachment of functional ligands to their carbon cage.

(*vii*) They can be used as *tracers molecules* by trapping lanthanide metals inside their cage.

(*viii*) They are useful in *fighting the HIV virus* that leads to AIDS.

(*ix*) They are impervious to lasers and can be used as *defensive measures in war.*

3.5.4 Carbon Nanotubes (CNTs)

Carbon nanotubes are hollow cylinders of carbon atoms. Their appearance is like rolled tubes of graphite. The ends of CNTs have domed structures of six-membered rings capped by a five-membered ring. Carbon nanotubes (CNTs) were first isolated and characterized by Ijima in 1991.

There are two types of CNTs: Single-walled carbon nanotubes (SWNTs) and multi-walled carbon nanotubes (MWNTs). As their names imply, SWNTs consist of a single, cylindrical graphene layer, where as MWNTs consist of multiple graphene layers fused into one another. A large number of articles have been published on CNTs and many new applications of CNTs are emerging every year.

Characteristics of Carbon nanotubes

The unique physical and chemical properties of CNTs, such as structural rigidity and flexibility has increased their utility. Their important characteristics are given below:

(*i*) CNTs are extremely strong, about 100 times stronger than steel.

(*ii*) CNTs can also act as either conductors of semiconductors depending on their chirality.

(*iii*) They can act as superconductors, and can also behave as field emitters.

(*iv*) They possess very low density about half of density of aluminium metal.

(*v*) They possess high elasticity and high thermal conductivity almost twice as that of diamond.

(*vi*) They have high current density and high aspect ratio *i.e.*, length to diameter ratio.

Applications

(*i*) Carbon nanotubes are now the top candidates to replace *silicon based semiconductors* as they have variable semiconducting properties with energy gaps ranging from a few milli electron volt (meV) to a few tenths of an electron volt (eV).

(*ii*) Nanotubes have led to a new generation of *electronic devices.*

(*iii*) Their use as ultra-sensitive *electromechanical sensors* has also been explored.

(*iv*) They are used in designing of *field-effect transistors (FETs) and integrated electronic devices.*

(*v*) They are used in *designing chemical and radiation sensors.*

(*vi*) They are used in composites and *manufacture of fuel cells.*

Scanned with OKEN Scanner

3.5.5 Dendrimers (Organic Nanoparticles)

Dendrimers are macromolecules having special structure and can be classified as a special type f polymers. These nanometer sized, polymeric systems are hyperbranched materials. The surface of endrimers has many chain ends which can be utilized to perform different chemical functions. They ıay be water-soluble but, because of their compact dimensions, they do not have the usual thickening roperties that many polymers have in solution. Dendrimers are composed of three components:

(*i*) A central core

(*ii*) An interior dendritic structure (the branches), and

(*iii*) An exterior surface (the end groups).

Over 50 families different in composition and structure of these nanoscale macromolecules are nown.

Characteristics of Dendrimers

(*i*) They are characterized by nearly spherical structures of nanometer sizes.

(*ii*) They have large numbers of reactive endgroups.

(*iii*) They contain interior cavities into which other molecules could be placed and are thus used in drug delivery.

(*iv*) They have low toxicity.

Applications of Dendrimers

(*i*) Due to their specific properties they are used in nanotechnology applications in both *biological and materials sciences*.

(*ii*) They find applications in many fields, including material engineering, industrial, pharmaceutical, and biomedical applications.

(*iii*) They are used as *nanoscale catalysts and novel lithographic materials.*

(*iv*) They have been used in many fields such as targeted *drug delivery systems, magnetic resonance imagine* (as contrast agents), and *bioadhesives.*

13.5.6 Nanocomposites

Nanocomposites are materials with a nanoscale structure that improve the macroscopic properties of products. Typically, nanocomposites are clay, polymer or carbon, or a combination of these materials with nanoparticle building blocks. Nanocomposites, materials with nanoscale separation of phases can generally be divided into two types:

(*i*) *multilayer structures* which are formed by gas phase deposition or from the self-assembly of monolayers.

(*ii*) *inorganic/organic composites* which can be formed by sol-gel techniques, bridging between clusters, or by coating nanoparticles, in polymer layers.

Applications of Nanocomposites

(*i*) Nanocomposites can greatly enhance the properties of materials. For example, ppm level impurities in aluminum alloys, increase their strength and corrosion resistance.

(*ii*) Magnetic multilayered materials are one of the most important aspects of nanocomposites as they have led to significant advances in storage media.

Scanned with OKEN Scanner

13.5.7 Biological Nanomaterials

The materials of biological origin are referred to as biological materials. These are used for nanotechnogical applications. These materials comprised of biological molecules and have specific application which belong to proteins, nucleic acids, lipids, hormones, vitamins etc. Biological nanomaterials show properties such as: Self-assembling properties and specific molecular recognition.

Applications of Biological Nanomaterials

(*i*) Biological nanomaterials are used for specific filtration and drug delivery.
(*ii*) They are used as sensors, motors and actuators.
(*iii*) They are also used in arrays (chips) for large scale protein or DNA screening.
(*iv*) These materials find application in the formation of memory devices.

13.5.8 Nanowires

Nanowires are ultrafine wires or linear arrays of dots, formed by self-assembly. They can be made from a wide range of materials. Semiconductor nanowires made of silicon, gallium nitride and indium phosphide have demonstrated remarkable optical, electronics and magnetic characteristics.

Applications of Nanowires

(*i*) Nanowires have potential applications in *high-density data storage, as magnetic and electronic nanodevices.*
(*ii*) The nanowires can be used in *Chemical Vapour Deposition (CVD) technique.*

13.6 FUTURE PROSPECTS OF NANOTECHNOLOGY

The impact of nanotechnology is expanding and nothing will remain untouched by this field in the near future. Its applications are enormous and limitless. Nanotechnology enables us to do things better than conventional technology. Nanotechnology is sure to revolutionise the most demanding fields of the present age namely:

1. Computer and electronic 2. Medicines 3. Pollution control

For example, there are enormus applications in computers to increase memory and speed. It is said that computers will run fast, medicines will cure all diseases, and pollution could be eliminated with the application of nanotechnology. The applications are immense, and it is predicted that nanotechnology will revolutionize the industrial world in the 21st century. The following are the major thrust areas, which are being developed by nanochemistry, for the benefit of future generation.

(*i*) Surface coatings consisting of nanomaterials become scratch proof, weather proof, more durable, better shining and dirt repellant.
(*ii*) Nanotechnology can enable drugs to destroy cancer or tumor without any side-effects.
(*iii*) Nanotubes pacemakers placed in the heart blood vessels could generate electricity for heart function.
(*iv*) Nanosized robots can repair damaged and diseased tissues.
(*v*) Converting sunlight into power, cleaning the pond water for drinking.
(*vi*) Creating sensor in the form of biochips to tbe planted in human body.

Scanned with OKEN Scanner

i) Nanosized titanium dioxide is used in cosmetic, sunscreens, creams, and lotions, which become more durable.

i) Nano impregnated clothes can block chemical and biological weapons from touching the skin.

:) Nanotechnology can help to detect narcotics and finger prints of criminals.

:) Nano assembled monolayer could be coated on to the fibres of the cloths so that it does not attract dirt and becomes self-cleaning and germ free.

he future prospects of nanotechnology are very bright, as it directly concerns the quality, comfort er health of the masses. Lot of academic institutions, Research and Development labs and :s are likely to come up. Government of India is promoting focused programmes in nology and huge funds are being allocated to scientific departments for growth of nanosciences

INFORMATION WINDOW

nemistry

he term nano means one billionth or 10^{-9} of a meter.

ano chemistry is the study of fundamental principles of molecules and structures with at least ne dimension roughly between 1 and 100 nanometers.

ano scale is defined as to be from 100 nm down to the size of atoms because of thin scale roperties of material can be different from those of a larger scale.

ue to increased relative surface area and quantum effects the properties of nanomaterials change rastically.

anochemistry is also utilization of synthetic chemistry building blocks, of different size and iape, composition and surface structure, charge and functionality.

anochemistry finds applications in almost all branches of science and engineering for e.g., in brics, strength of materials, drug delivery, micro/nano electromechanical system, molecular anufacturing etc.

Top down" approach is fabricating nano-objects from large entities without atomic level control.

Bottom up" approach is used for synthesis in which materials and devices are build from olecular component.

ano-crystals are crystals of nanometer dimensions (of 1 to 50 nm) consisting of aggregates of few hundred to ten thousands of atoms combined to a cluster.

ano-crystals finds application in high density data storage, as efficient light emitters.

ue to increased surface area as compared to macro-scale nano-metals and oxides are ultra-high tivity catalyst.

anometals and oxides are widely used in the formation of nanocomposites.

ullerenes are carbon allotropes. Spherical fullerenes are also called bucky balls and cylindrical es are called carbon nanotubes or bucky tubes.

ullerenes are defined as an elongated sphere of sp^2 carbon atoms, densely packed in a honeycomb ystal lattice.

rst isolated and characterized fullerence C_{60}, contains 20 hexagonal faces and 12 pentagonal ces.

cky ball clusters are smallest members with 20 carbons. Nanotubes are hollow tubes of very all dimensions having single or multiple walls and have potential applications in electronics lustry.

Scanned with OKEN Scanner

13.5.2 Nanocrystalline Materials

These include ceramics, metals, and metal oxide nanoparticles. In the last two decades a class of materials with a nanometer-sized microstructure have been synthesized and studied. These materials are assembled from nanometer-sized building blocks, which are mostly crystallites. The building blocks may differ in their atomic structure, crystallographic orientation, or chemical composition.

Applications

(*i*) Due to their increased surface area as compared to macro-scale materials, nanometals and oxides are ultra-high activity catalysts.

(*ii*) They are also used as desirable starting materials for a variety of reactions, especially solid-state reaction.

(*iii*) Nanometals and oxides are also widely used in the formation of nanocomposites.

(*iv*) They have many useful and unique magnetic, electric, and optical properties.

13.5.3 Fullerenes (C_{60})

Fullerenes are family of carbon allotropes, *i.e.*, molecules composed of entirely carbon, in the form of a hollow spare, ellipsoid, tube or plane. Spherical fullerenes are also called **buckyballs**, and cylindrical ones are called carbon nanotubes or **buckytubes**. **Graphene** is an example of a planar fullerenes sheet. Its structure is one atom thick planner sheets of Sp^2-bonded C atoms that are densely packed in a honeycomb crystal lattice. These include the carbon nanotubes (or silicon nanotubes), which are of interest both because of their mechanical strength and also because of their electrical properties.

Fullerenes are similar in structure to graphite, which is composed of sheets of hexagonal rings, but may also contain pentagon rings.

Fullerenes are defined as an elongated sphere of carbon atoms formed by interconnecting six-member rings and isolated five-member rings forming hexagonal and pentagonal faces. The first isolated and characterized fullerene, C_{60}, contains 20 hexagonal faces and 12 pentagonal faces just like a soccer ball and possess perfect symmetry.

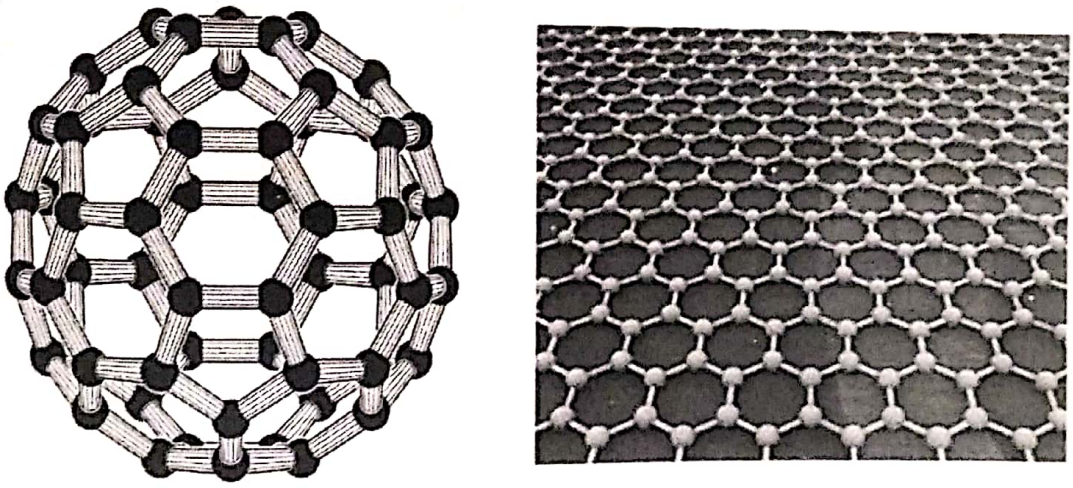


Fig. 13.2 (a) Buckminsterfullerene C_{60} (b) Graphene lattice

Scanned with OKEN Scanner

evelopment of different varieties of Fullerenes

Since the discovery of fullerenes in 1985, structural variations on fullerenes have led to the :velopment of many variety such as:

(*i*) **Buckyball clusters:** They are the smallest members with 20 carbons, comparable to unsaturated dodecahedrane and the most common is C-20.

(*ii*) **Nanotubes:** They are hollow tubes of very small dimensions, having single or multiple walls and have potential applications in electronics industry.

(*iii*) **Megatubes:** They are larger in diameter than nanotubes and have walls of different thickness. These are potentially used for the transport of a variety of molecules of different sizes.

(*iv*) **Polymers:** Two-dimensional and three-dimensional polymers are formed under high pressure and high temperature conditions.

(*v*) **Nano "opions":** They are spherical particles based on multiple carbon layers surrounding a buckyball core. They are used as lubricants.

(*vi*) **Hydrated fullerene (HyFn):** Hydrated fullerene C_{60}HyFn is a stable, highly hydrophilic, supra-molecular complex consisting of C_{60} fullerene molecule enclosed into the first hydrated shell that contains 24 water molecules: $C_{60}@(H_2O)_{24}$. This hydrated shell is formed as a result of donor-acceptor interaction between lone-electron pairs of oxygen, water molecules and electron acceptor centres on the fullerene surface. Meanwhile, the water molecules which are oriented close to the fullerene surface are interconnected by a three-dimensional network of hydrogen bonds. The size of C_{60}HyFn is 1.6-1.8 nm. The maximal concentration of C_{60} in the form of C_{60}HyFn achieved by 2010 is 4 mg/mL.

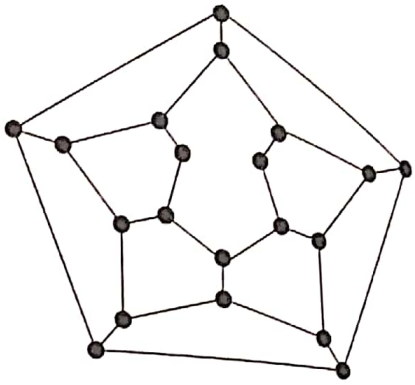


Fig. 13.3 C_{60}HyFn water solution with a C_{60} concentration of 0.22 g/L

(*i*) **Fullerite:** Fullerites are the solid-state manifestation of fullerenes and related compounds and materials. "Ultrahard fullerite" is a coined term frequently used to describe material produced by high-pressure high-temperature (HPHT) processing of fullerite. Such treatment converts fullerite into a nanocrystalline form of diamond which has been reported to exhibit remarkable mechanical properties.

racteristics of Fullerenes

) They are insoluble in water, but soluble in organic solvents. C-60 is soluble in benzene and can be crystallized from it.

) They exhibit excellent tensile strength and very high packing density.

) C-60 is non-toxic and a powerful antioxidant. In living cells, it reacts with free radicals responsible for the damage.

They possesses cage like structure.

They are impermeable to all elements under normal circumstances.

C-60 crystal is an insulator but when dopped with an alkali atom, it becomes conducting.

C-60 crystal doped with potassium becomes superconductor at 15 K.

Scanned with OKEN Scanner

Characteristics that a structure should possess by any of the fabrication techniques

a) Identical small size of all particles (also called mono sized or with uniform size distribution.
b) Identical shape or morphology.
c) Identical chemical composition and crystal structure that are desired among different particles and within individual particles, such as core and composition must be the same.
d) Particles should be individually dispersed or mono dispersed, *i.e.*, no agglomeration.

Now let us understand some of the important concepts of nano technology:

NANOSCALE MATERIALS

Nanoscale materials are materials with a microstructure, the characteristic length scale of which e order of a few nanometers (1-100). The microstructure refers to the chemical composition, the ment of the atoms, and the size of a solid in one, two, or three dimensions.

Physical electronic or catalytic properties of solids are altered if there is interaction at nano s mentioned earlier, materials reduced to the nanoscale can suddenly show very different properties ed to what they exhibit on a macroscale, enabling unique applications. For instance, opaque ices become transparent (copper); inert materials attain catalytic properties (platinum); stable ils turn combustible (aluminum); solids turn into liquids at room temperature (gold); insulators e conductors (silicon). Materials such as gold, which is chemically inert at normal scales, can s a potent chemical catalyst at nanoscales. Much of the fascination with nanotechnology stems ese unique quantum and surface phenomena that matter exhibits at the nanoscale.

Nanomaterials can be constructed by '**top down**' techniques, producing very small structures arger pieces of material. For example, to create circuits on the surface of a silicon microchip by g. They may also be constructed by '**bottom up**' techniques, which means atom by atom or ule by molecule.

Although, we categories nonmaterials as those which have structured components with at least mension less than 100 nm. Materials that have one dimension in the nanoscale and are extended other two dimensions are layers, such as a thin films or surface coatings. Materials that are cale in two dimensions (and extended in one dimensions include nanowires and nanotubes.

Materials that are nanoscale in three dimensions are particles. For example, precipitates, colloids uantum dots tiny particles of semiconductor materials. Nanocrystalline materials, made up of netre-sized grains, also fall into this category.

The various classes of nanoparticles that serve as the building blocks of nanomaterials and devices been discussed below.

.1 Nanocrystals and Clusters

Nanocrystals are crystals of nanometer dimensions, usually consisting of aggregates of a few ed to ten thousands of atom combined into a cluster. Nanocrystals have typical dimensions of 1 to nometers (nm), and thus they are intermediate in size between molecules and bulk materials and it properties that are also intermediate.

ications

Nanocrystals are of great interest because of their promise in high density data storage and in electronic applications, as they can be efficient light emitters.

Nanocrystals have also found applications as biochemical tags, as laser and optical components, for the preparation of display devices, and

For chemical catalysis.

Scanned with OKEN Scanner